

Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – VI

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Class: SYBSc IT	Subject: DATA STRUCTURE
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PART – B : Question

Tree Traversal: Write a program to:

a. Implement pre-order

CODE:

```
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class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def preorder(root):
    if root:
        print(str(root.val) + "-->", end="")
        preorder(root.left)
        preorder(root.right)

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("Preorder Traversal")
preorder(root)
print("\nIQRA S021")
|
```

OUTPUT:

```
===== RESTART:
Preorder Traversal
1-->2-->4-->5-->3-->
IQRA SO21
|
```

b. In-order

CODE:

```
class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def inorder(root):
    if root:
        inorder(root.left)
        print(str(root.val) + "-->", end="")
        inorder(root.right)

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("Inorder Traversal")
inorder(root)
print("\nIQRA SO21")
|
```

OUTPUT:

```
===== RESTART:
Inorder Traversal
4-->2-->5-->1-->3-->
IQRA SO21
|
```

c. Post-order traversal of a binary tree.

CODE:

```

class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def postorder(root):
    if root:
        postorder(root.left)
        postorder(root.right)
        print(str(root.val) + "-->", end="")

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("Postorder Traversal")
postorder(root)
print("\nIQRA SO21")
|

```

OUTPUT:

```

===== RESTART: D
Postorder Traversal
4-->5-->2-->3-->1-->
IQRA SO21

```